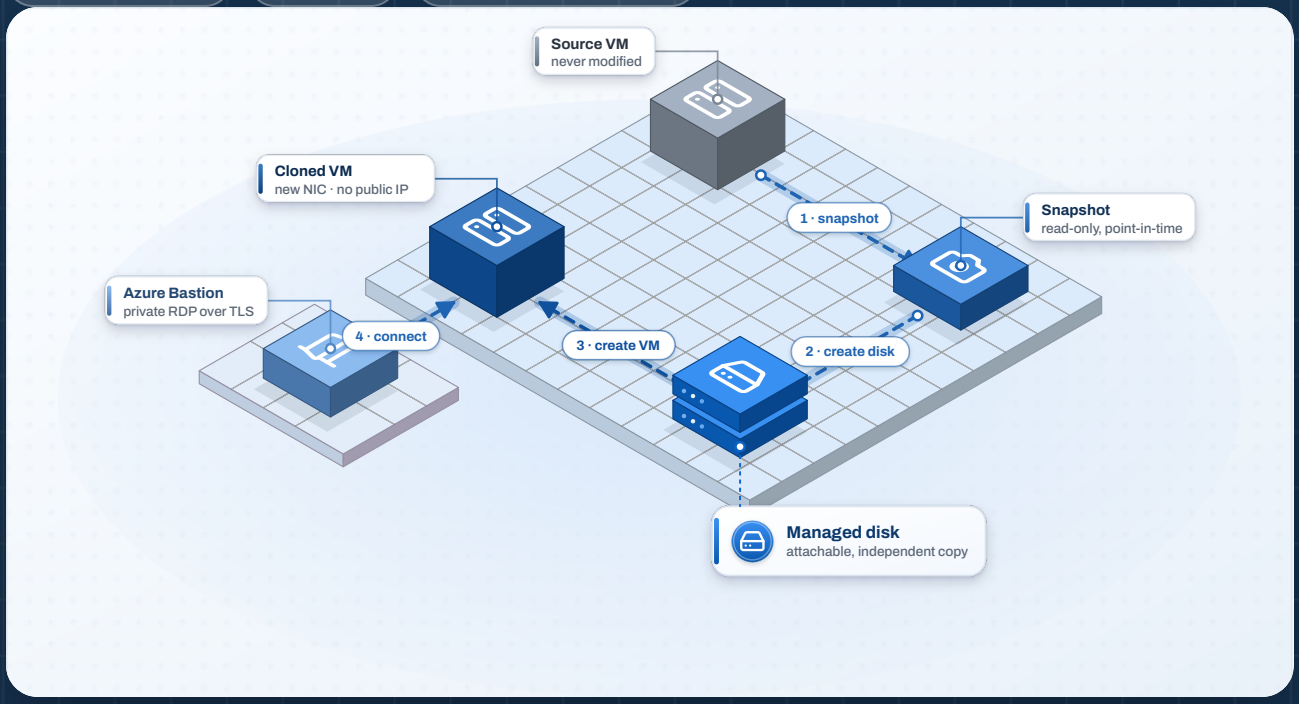
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-AZ-003

Cloning an Azure Virtual Machine

Snapshot → managed disk → new VM, done four ways (portal, PowerShell, Azure CLI, Compute Gallery), with secure access and full rollback.

Microsoft Azure | Compute | Change Management



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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain why Azure has no single “Clone” button and what a snapshot, a managed disk and an OS disk are.
- Choose between a specialized clone, a generalized gallery image and a backup restore for a given business need.
- Clone a VM through the Azure portal, with an Az PowerShell template, and with the Azure CLI.
- Identify the least-privilege Azure RBAC roles required at each stage.
- Connect to the clone without a public IP using Azure Bastion or Just-In-Time access, then rename and harden it.
- Roll back cleanly at any stage and produce audit evidence for the change record.

TIME	DIFFICULTY	MINIMUM ROLE	RISK TO SOURCE VM
≈ 40 min active + 10 min deploy	Intermediate	Virtual Machine Contributor + Disk Snapshot Contributor (RG)	None (read-only copy)

CHAPTER 1

Concepts You Need First

Cloning a virtual machine sounds like it should be a single click. In Microsoft Azure it is not, and the reason tells you a lot about how Azure stores machines. A *virtual machine* in Azure is not one object. It is a small bundle of separate resources: a compute definition (size, zone, security type), one or more **managed disks**, a **network interface (NIC)**, and optionally a public IP address. When you "clone" a VM, you are really copying its *disk* and then building a new compute definition and NIC around that copy.

> **KEY CONCEPT · A VM IS A RECIPE, THE DISK IS THE MEAL**
>
> Think of the VM resource as a recipe card: how many vCPUs, how much memory, which network. The OS disk is the cooked meal: the operating system, installed software, user profiles and data. To make a second, identical meal you don't photocopy the recipe card. You copy the meal and write a new card for it. That is exactly the snapshot → disk → VM pattern.

1.1 The three building blocks

Resource	What it is	Can you boot from it?
Snapshot	A read-only, point-in-time copy of one managed disk, stored as its own resource. *Full* snapshots hold a complete copy; *incremental* snapshots hold only the blocks changed since the last snapshot and are cheaper.	No. A snapshot cannot be attached to a VM.
Managed disk	A block-storage volume that Azure manages for you (Standard HDD, Standard SSD, Premium SSD, Premium SSD v2 or Ultra). It can be created *from* a snapshot.	Yes, when attached as the OS disk of a VM.
Virtual machine	The compute definition plus its NIC, attached disks, identity, extensions and tags.	It is the thing that boots.

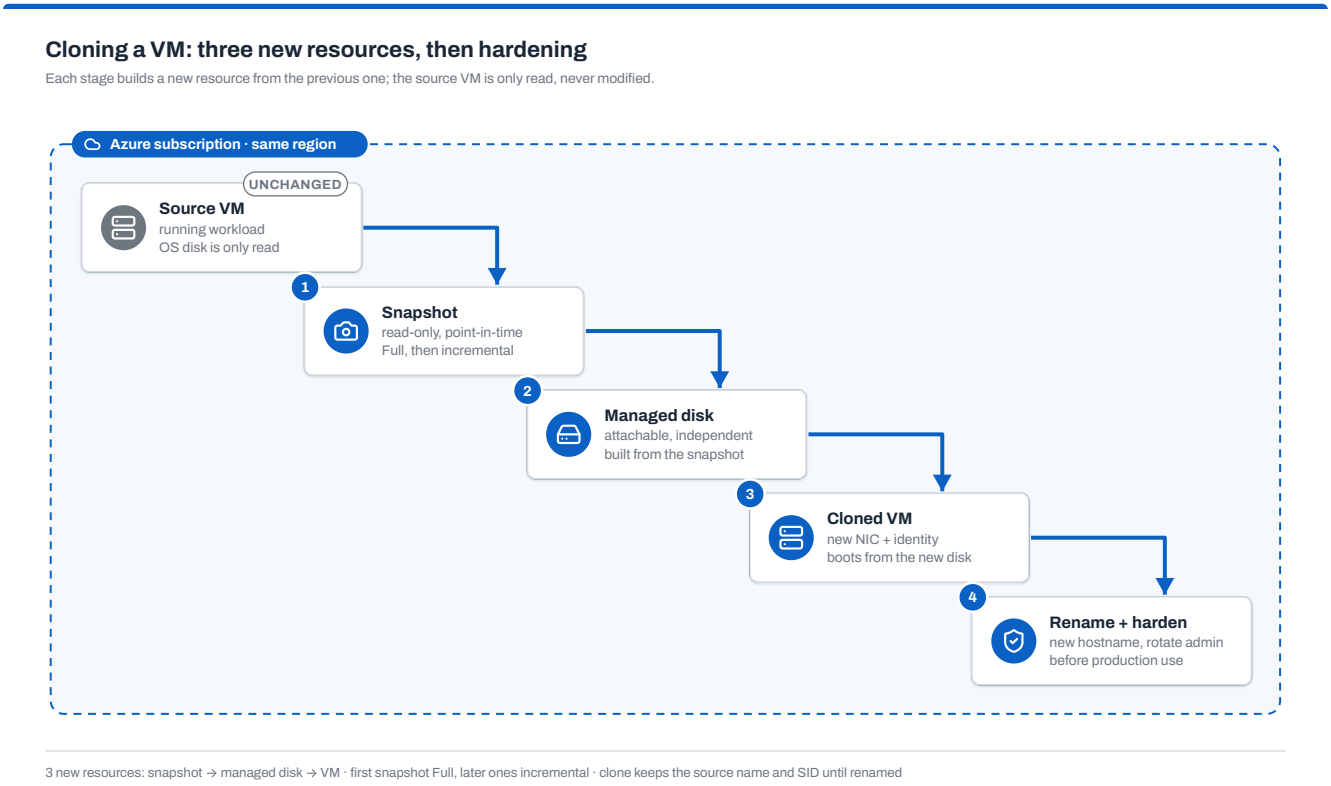


Figure 1. The source VM is only ever read, so a failed clone at any stage is cleaned up by deleting what you created, never by repairing the source.

1.2 Specialized versus generalized copies

There are two fundamentally different kinds of disk copy, and choosing the wrong one is the most common cloning mistake.

- **Specialized**: a byte-for-byte copy. The clone keeps the source's computer name, local accounts, Windows security identifier (SID), installed apps and domain trust secret. This SOP's main methods (A, B and C) create specialized clones. They are ideal for *test copies, upgrade rehearsals and forensic copies*.
- **Generalized**: the OS has been run through `sysprep /generalize` (Windows) or `waagent -deprovision` (Linux), which strips machine-specific identity. Every VM built from it gets a fresh name and SID. Use generalized images when you need *many* VMs from one golden build (Method D).

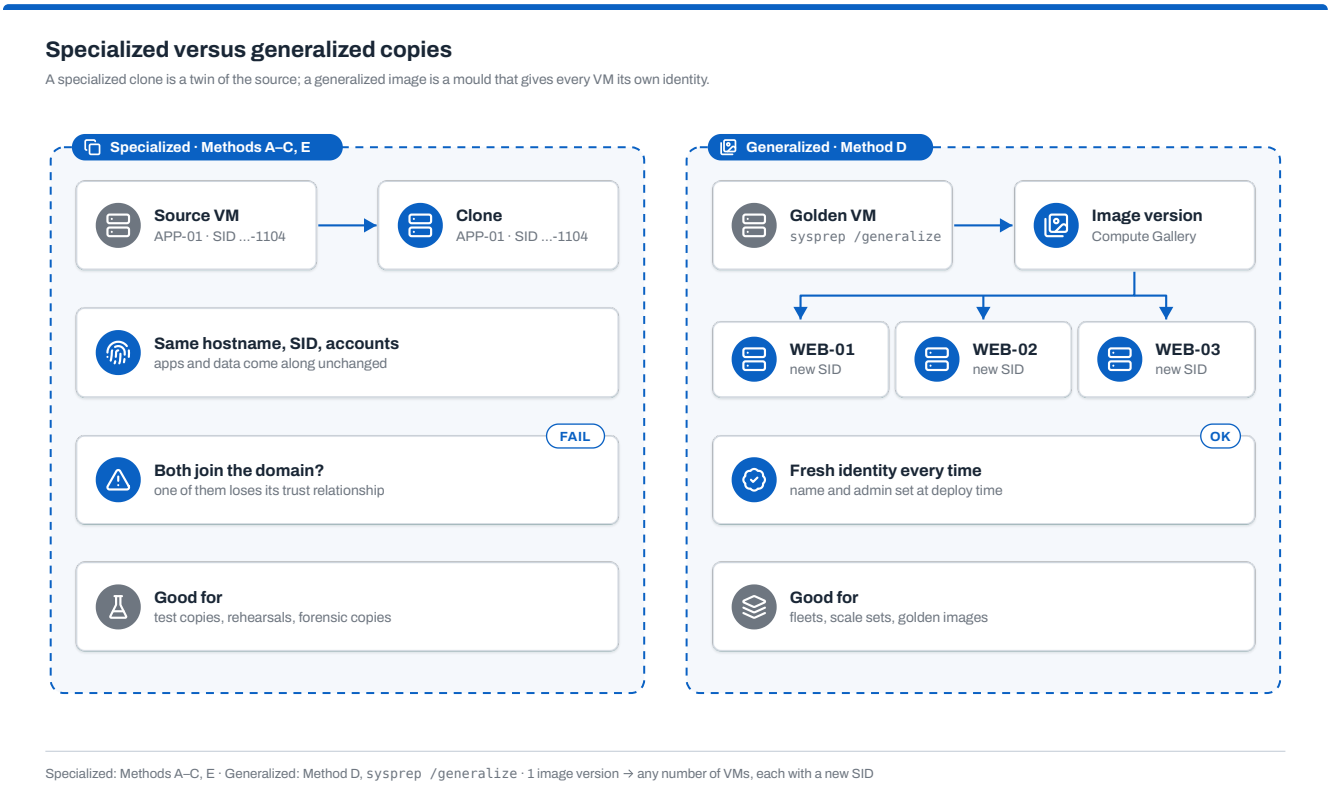


Figure 2. A twin is what a test rehearsal or a forensic copy needs and exactly what a domain cannot tolerate twice; a mould gives every VM its own identity.

> **WATCH OUT · DUPLICATE IDENTITIES**
>
> A specialized clone boots with the **same hostname and SID** as the source. If both machines join the same Active Directory domain at once, one of them will lose its secure channel with the domain controller ("The trust relationship between this workstation and the primary domain failed"). Keep the clone on an isolated network or rename it *before* it reaches the domain (Chapter 6).

1.3 Crash-consistent versus application-consistent

A snapshot of a *running* VM is **crash-consistent**: it is equivalent to pulling the power cord. Files are intact, but writes that were still in memory (for example an open SQL transaction) may be missing. Stopping (deallocating) the VM first gives you a clean, fully flushed disk. For production databases that cannot be stopped, use Azure Backup or VM restore points, which coordinate with the Volume Shadow Copy Service (VSS) to produce **application-consistent** copies (Method E).

1.4 Choosing a method

A · Azure portal	B · Az PowerShell	C · Azure CLI	D · Compute Gallery image	E · Backup / restore point
Point-and-click, three wizards. Best for learning and one-off clones.	Parameterized script that runs all three stages and tags everything.	Same flow as B in Bash. Runs in Cloud Shell or any Linux/macOS terminal.	Capture a generalized image, then deploy as many VMs as needed.	Restore a past, application-consistent state into a new VM.
BEST FOR: AD-HOC WORK	BEST FOR: REPEATABLE, AUDITED CHANGES	BEST FOR: PIPELINES AND LINUX ADMINS	BEST FOR: FLEETS AND GOLDEN IMAGES	BEST FOR: DR TESTS, "CLONE AS OF LAST NIGHT"

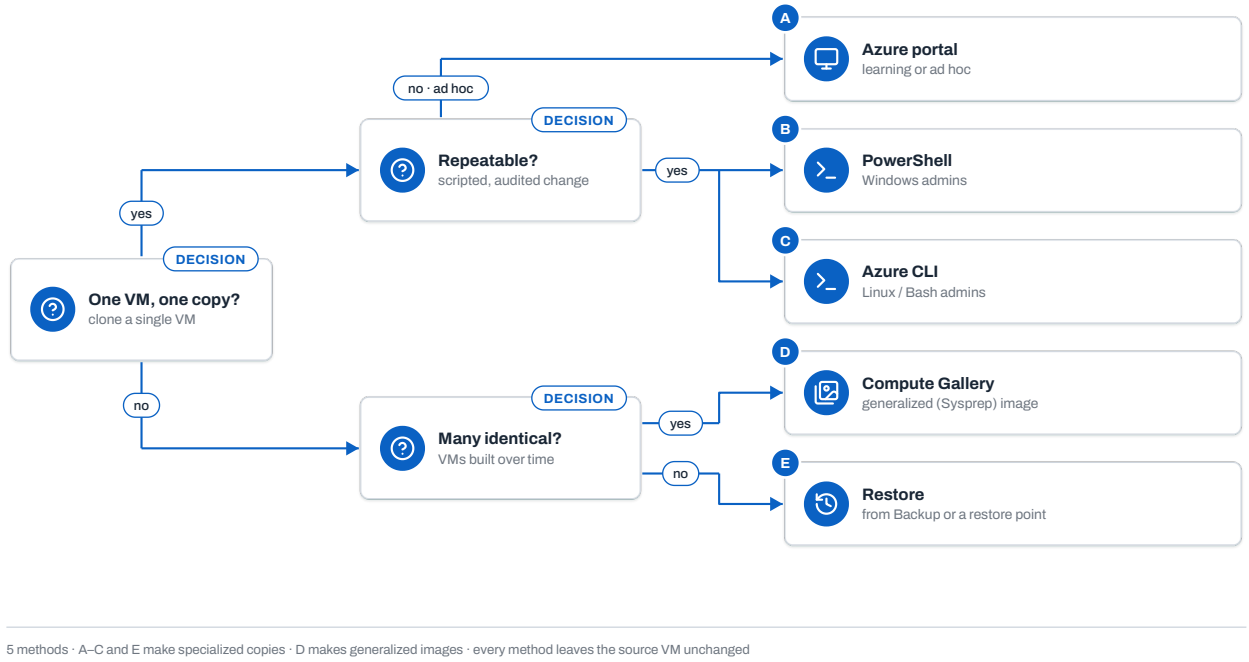


Figure 3. Most one-off clones are Methods A to C; reach for D when you will build the same VM again and E when you need yesterday rather than today.

CHAPTER 2

Access, Prerequisites and Planning

2.1 Level of access required

Azure uses **role-based access control (RBAC)**. A role is a named bundle of permissions, and it is granted at a *scope*: management group, subscription, resource group or individual resource. Permissions granted at a higher scope flow down to everything beneath it. Following the principle of least privilege, grant roles at the **resource group** that will hold the clone, not at the subscription.

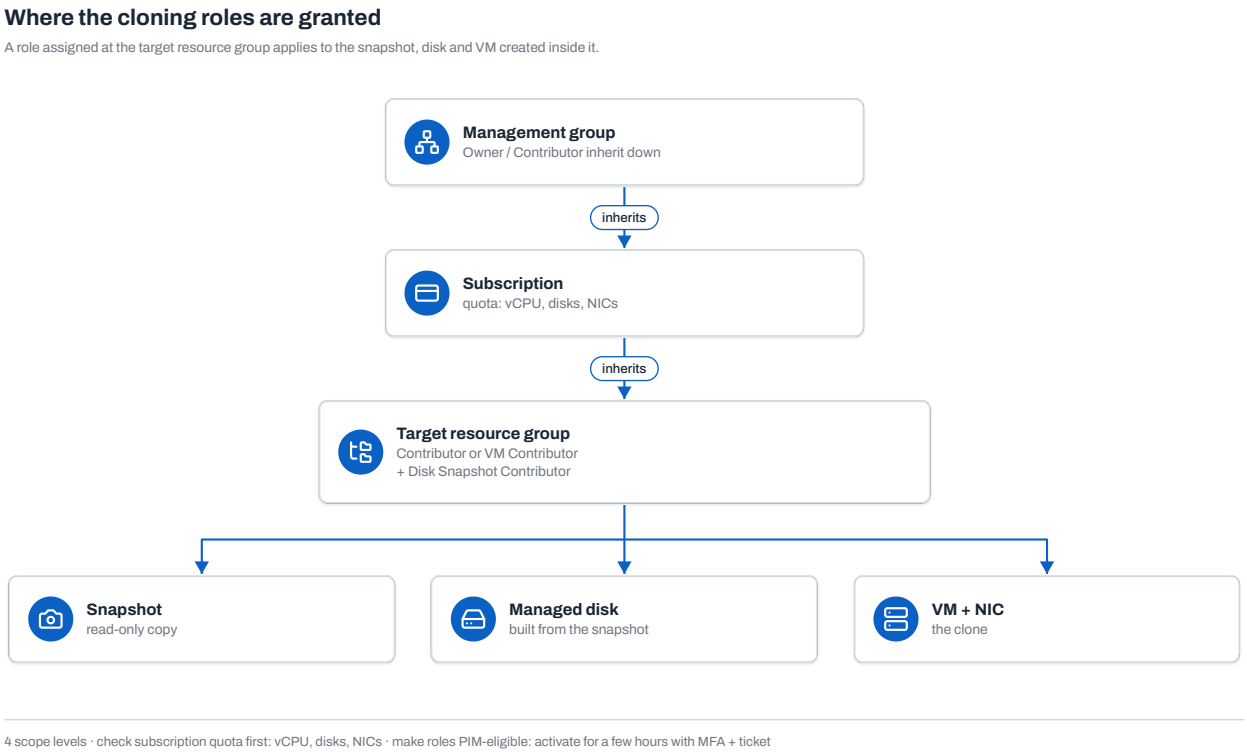


Figure 4. Assign the roles at the target resource group and they cover every snapshot, disk and VM the clone creates there.

Task	Least-privilege built-in role	Notes
Read the source VM and its disk	Reader on the source resource group	Needed to select the OS disk.
Create the snapshot	Disk Snapshot Contributor	Also requires `Microsoft.Compute/disks/beginGetAccess/action` on the source disk, which Contributor provides.
Create the managed disk	Virtual Machine Contributor or Contributor	VM Contributor can manage disks attached to VMs.
Create VM + NIC	Virtual Machine Contributor *and* Network Contributor (or Contributor)	The NIC joins an existing subnet, which needs `Microsoft.Network/virtualNetworks/subnets/join/action`.
Stop/deallocate the source	Virtual Machine Contributor	Optional but recommended.
Connect with Bastion	Reader on the VM, NIC and Bastion resource	Plus *Virtual Machine Administrator Login* if using Entra ID sign-in.
Request JIT access	Reader + `Microsoft.Security/locations/jitNetworkAccessPolicies/initiate/action`	Defender for Servers Plan 2 must be enabled.

GOOD PRACTICE · JUST-IN-TIME PRIVILEGE

Make these roles *eligible* rather than permanent with Microsoft Entra Privileged Identity Management (PIM). The administrator activates Contributor for a few hours, with MFA and a ticket number, and the activation itself becomes audit evidence.

2.2 Prerequisites checklist

- [ ] Approved change request; its number will be applied as a `ChangeRef` tag on every new resource.
- [ ] Quota in the target region for one more VM of the chosen size (check **Subscriptions › Usage + quotas**).
- [ ] The source VM's local administrator credentials (the clone inherits them).
- [ ] Azure Bastion in the target virtual network, or Defender for Cloud JIT enabled.
- [ ] If the source disk uses a customer-managed key (CMK), the clone's region must reach the same Key Vault and disk encryption set.

2.3 Decisions to make up front

Decision	Guidance
Clone name	`<source>-CLONE` or `<source>-TEST`. Must be unique in the resource group.
Network placement	Isolated VNet for test copies of domain members; same VNet only after rename.
Size and disk tier	Match the source for performance testing; use B-series and Standard SSD to save money.
Snapshot type	Full for the first snapshot of a disk; incremental thereafter.
Cleanup	Enable **Delete with VM** on disk and NIC so teardown is one action.

CHAPTER 3

Method A · Cloning in the Azure Portal

This is the visual walk-through. Each stage happens in a different blade of the portal, and each one ends with **Review + create**.

STAGE 1 Create the snapshot

1. Sign in to **portal.azure.com** with MFA and activate your PIM role.
2. Open **Virtual machines › <source VM>**. Recommended: select **Stop** and wait for *Stopped (deallocated)*.
3. Under **Settings › Disks**, select the OS disk name to open the disk blade.
4. Select **+ Create snapshot** and complete the form: name `<VM>-SNAPSHOT`, snapshot type **Full**, storage type Standard HDD (cheapest; performance does not matter for a snapshot), tags including `ChangeRef`.
5. Select **Review + create** › **Create**. After 1–3 minutes, confirm *Provisioning state: Succeeded* and *Completion percent: 100*.

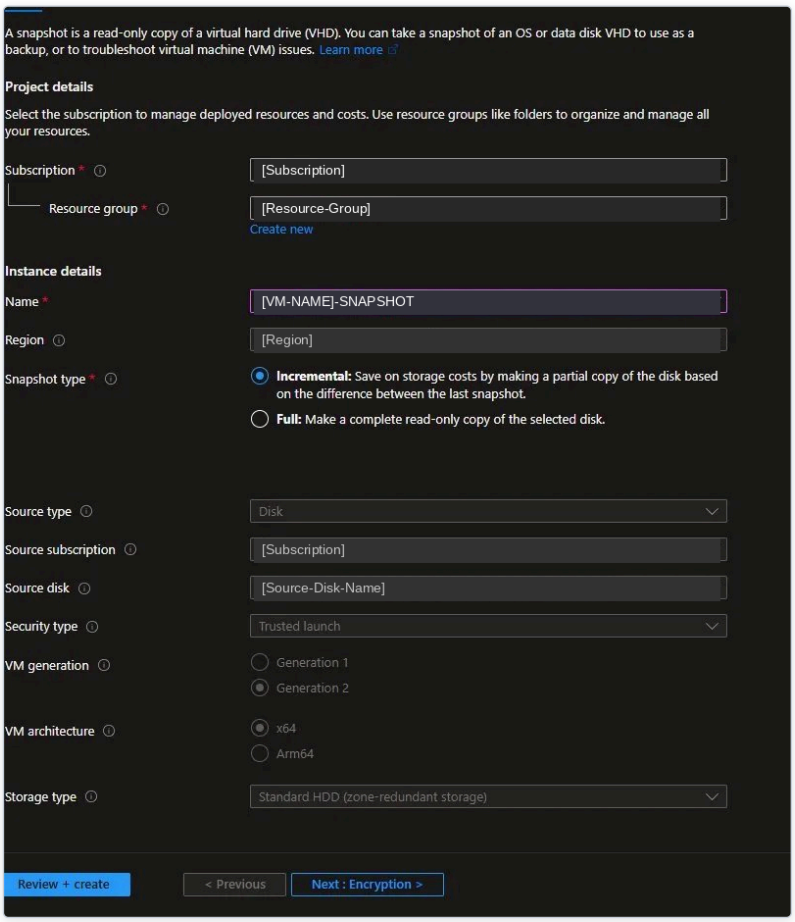


Figure 5. The Create snapshot form. Change *Snapshot type* to **Full** for the first snapshot of a disk.

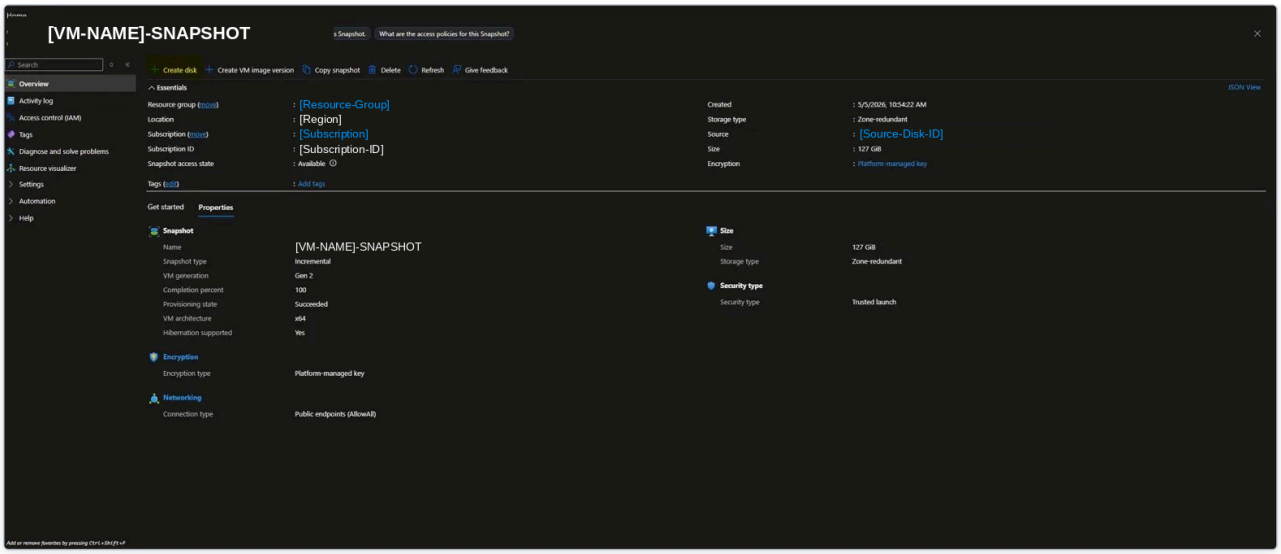


Figure 6. Snapshot overview showing Provisioning state Succeeded. Note the `Create disk` button in the toolbar, which is the entry point to Stage 2.

> **NOTE · DATA DISKS**
>
> A snapshot covers *one* disk. If the source has data disks, repeat Stage 1 for each (`<VM>-DATAsnap-1`, `-2` …) and attach the resulting disks in Stage 3.

STAGE 2 Create a managed disk from the snapshot

1. From the snapshot blade select **`+ Create disk`**.
2. Name it `<CLONE>-OSdisk`, keep the region, choose the availability zone the clone will use, and pick a storage tier (Premium SSD for production-like tests).
3. On the **`Networking`** tab set network access to **Disable public access** or a private endpoint. Disks can otherwise be exported via SAS URL.
4. Apply tags, then **`Review + create`**. Wait for *Validation passed* and select **`Create`**.

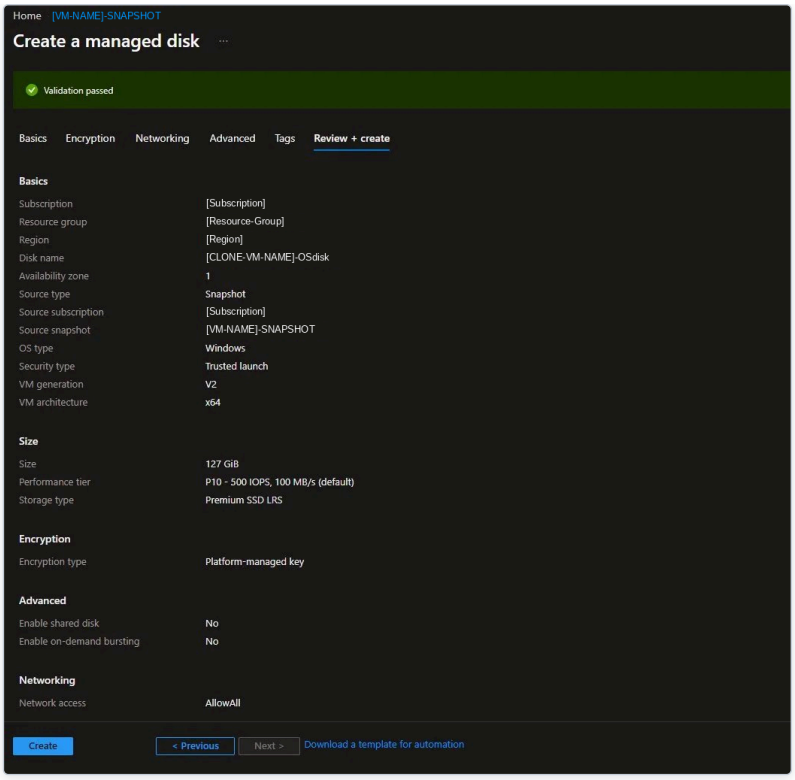


Figure 7. Review screen for the new managed disk. Source type is *Snapshot* and the OS type and security type are inherited.

STAGE 3 Create the VM from the disk

1. Open the new disk and select `+ Create VM`. The VM wizard opens with the *Image* field locked to your disk.
2. **Basics**: unique VM name, same region and zone as the disk, size as planned. The username and password fields are disabled because the clone inherits the source credentials. Set `Public inbound ports` to **None**.
3. **Disks**: enable `Delete with VM`; attach any cloned data disks with `Attach an existing disk`.
4. **Networking**: choose the planned VNet and subnet, set `Public IP` to **None**, and enable `Delete public IP and NIC when VM is deleted`.
5. **Management / Monitoring**: enable boot diagnostics (managed storage), periodic update assessment, and auto-shutdown for non-production clones.
6. **Tags**: `ChangeRef`, `Owner`, `Environment`, `Purpose`. Then `Review + create` › `Create`.

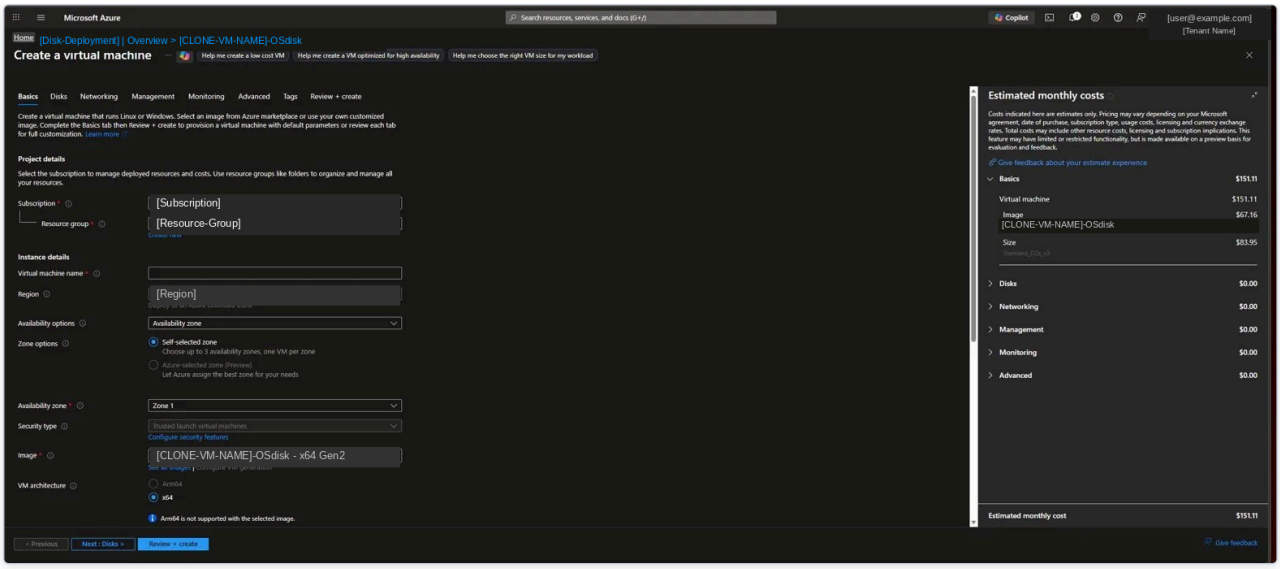


Figure 8. Basics tab. The Image is pre-set to the cloned OS disk, and the cost panel on the right estimates the monthly price.

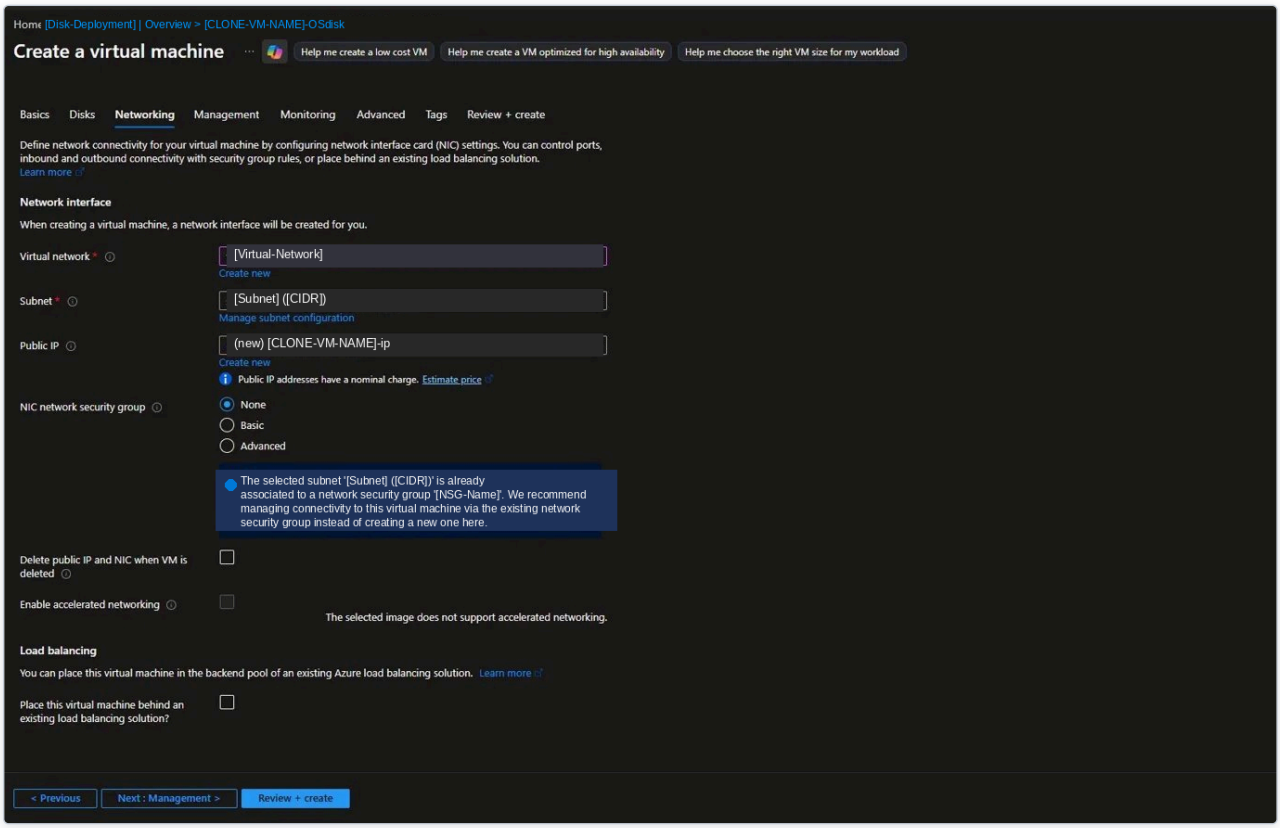


Figure 9. Networking tab. Public IP is *None*; when the subnet already has an NSG, leave the NIC NSG as *None* so the subnet rules govern access.

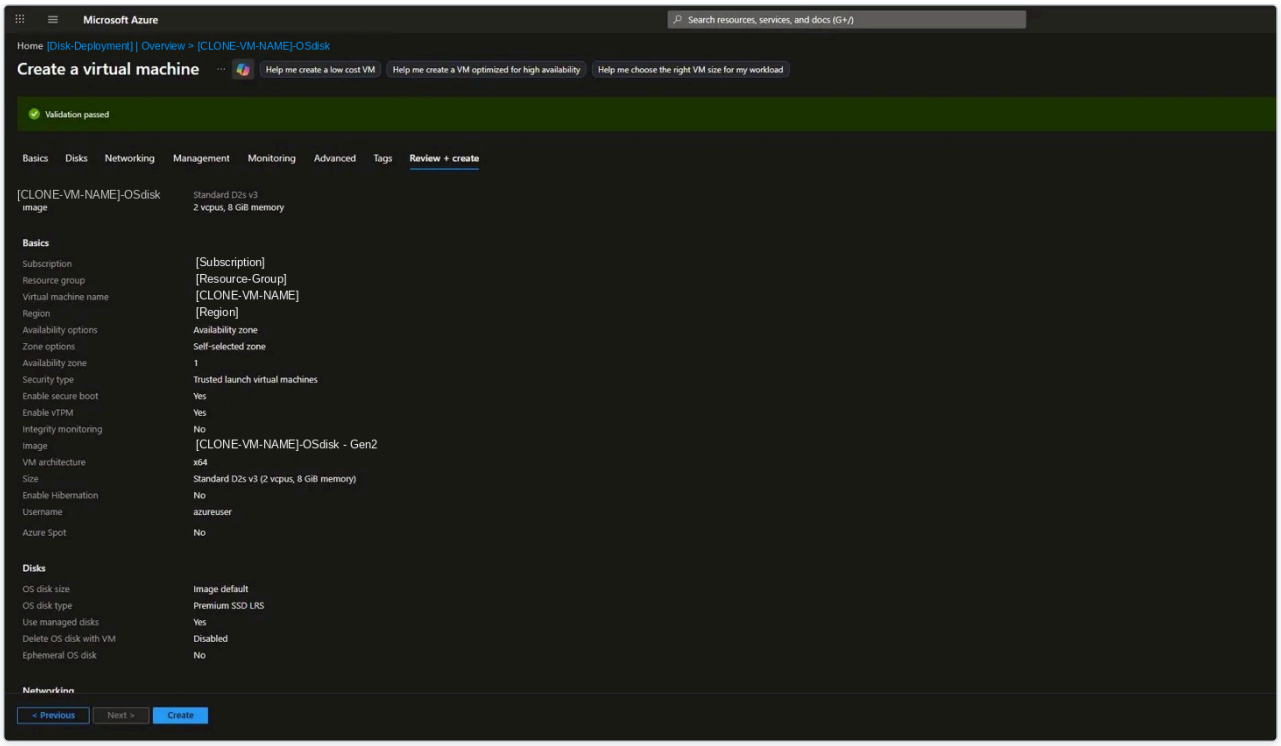


Figure 10. Review + create with *Validation passed*. The username shown is a placeholder; the clone signs in with the source VM's existing credentials.

CHAPTER 4

Method B · Az PowerShell Template

Scripting the clone makes it repeatable, peer-reviewable and self-documenting: the script *is* the change plan. The template below uses the **Az** PowerShell module (install with `Install-Module Az -Scope CurrentUser`), runs all three stages, applies tags everywhere, never creates a public IP, and writes a JSON evidence file for the change record.

CLONE-AZVM.PS1 · BASE TEMPLATE — POWERSHELL

```powershell
<#
.SYNOPSIS  Clone an Azure VM: snapshot -> managed disk -> new VM (no public IP).
.NOTES     Requires Az.Accounts, Az.Compute, Az.Network.  Run in PowerShell 7+ or Cloud Shell.
           Roles: Disk Snapshot Contributor + Virtual Machine Contributor + Network Contributor on the target RG.
#>
[CmdletBinding(SupportsShouldProcess)]
param(
  [Parameter(Mandatory)] [string] $SubscriptionId,
  [Parameter(Mandatory)] [string] $SourceResourceGroup,
  [Parameter(Mandatory)] [string] $SourceVmName,
  [Parameter(Mandatory)] [string] $TargetResourceGroup,
  [Parameter(Mandatory)] [string] $CloneVmName,
  [Parameter(Mandatory)] [string] $VnetResourceGroup,
  [Parameter(Mandatory)] [string] $VnetName,
  [Parameter(Mandatory)] [string] $SubnetName,
  [Parameter(Mandatory)] [string] $ChangeRef,              # e.g. CHG0012345
  [string] $VmSize      = '',                             # blank = match source
  [string] $DiskSku     = 'Premium_LRS',                  # Standard_LRS | StandardSSD_LRS | Premium_LRS
  [switch] $StopSource                                    # deallocate source for a clean snapshot
)
$ErrorActionPreference = 'Stop'
Connect-AzAccount -Subscription $SubscriptionId | Out-Null
$tags = @{ ChangeRef = $ChangeRef; Purpose = 'Clone'; Source = $SourceVmName; Owner = (Get-AzContext).Account.Id }

# ---- Stage 0: inspect the source ----------------------------------------
$src   = Get-AzVM -ResourceGroupName $SourceResourceGroup -Name $SourceVmName
$osRef = $src.StorageProfile.OsDisk
$osDisk = Get-AzDisk -ResourceGroupName $SourceResourceGroup -DiskName $osRef.Name
if (-not $VmSize) { $VmSize = $src.HardwareProfile.VmSize }
if ($StopSource -and $PSCmdlet.ShouldProcess($SourceVmName,'Stop-AzVM')) {
  Stop-AzVM -ResourceGroupName $SourceResourceGroup -Name $SourceVmName -Force | Out-Null
}

# ---- Stage 1: snapshot --------------------------------------------------
$snapCfg = New-AzSnapshotConfig -SourceUri $osDisk.Id -Location $osDisk.Location `
            -CreateOption Copy -SkuName Standard_LRS -Tag $tags
$snap = New-AzSnapshot -ResourceGroupName $TargetResourceGroup -SnapshotName "$SourceVmName-SNAPSHOT" -Snapshot $snapCfg

# ---- Stage 2: managed disk from snapshot (public network access disabled) -
$diskCfg = New-AzDiskConfig -Location $snap.Location -SourceResourceId $snap.Id -CreateOption Copy `
            -SkuName $DiskSku -Zone $src.Zones -OsType $osDisk.OsType -Tag $tags `
            -PublicNetworkAccess Disabled -NetworkAccessPolicy DenyAll
if ($osDisk.SecurityProfile) { $diskCfg.SecurityProfile = $osDisk.SecurityProfile }   # keep Trusted Launch
$newDisk = New-AzDisk -ResourceGroupName $TargetResourceGroup -DiskName "$CloneVmName-OSdisk" -Disk $diskCfg

# ---- Stage 3: NIC (no public IP) + VM attached to the disk --------------
$subnet = (Get-AzVirtualNetwork -ResourceGroupName $VnetResourceGroup -Name $VnetName).Subnets |
          Where-Object Name -eq $SubnetName
$nic = New-AzNetworkInterface -ResourceGroupName $TargetResourceGroup -Name "$CloneVmName-nic" `
         -Location $newDisk.Location -SubnetId $subnet.Id -Tag $tags
$vm = New-AzVMConfig -VMName $CloneVmName -VMSize $VmSize -Zone $src.Zones -Tags $tags
$vm = Add-AzVMNetworkInterface -VM $vm -Id $nic.Id -DeleteOption Delete
$isWin = $osDisk.OsType -eq 'Windows'
$vm = Set-AzVMOSDisk -VM $vm -ManagedDiskId $newDisk.Id -CreateOption Attach -DeleteOption Delete `
       -Windows:$isWin -Linux:(-not $isWin)
if ($src.SecurityProfile) {
  $vm = Set-AzVMSecurityProfile -VM $vm -SecurityType $src.SecurityProfile.SecurityType
  $vm = Set-AzVMUefi -VM $vm -EnableVtpm $true -EnableSecureBoot $true
}
$vm = Set-AzVMBootDiagnostic -VM $vm -Enable
New-AzVM -ResourceGroupName $TargetResourceGroup -Location $newDisk.Location -VM $vm | Out-Null

# ---- Evidence for the change record -------------------------------------
[pscustomobject]@{
  ChangeRef = $ChangeRef; Operator = (Get-AzContext).Account.Id; TimeUtc = (Get-Date).ToUniversalTime()
  SourceVm  = $src.Id; SnapshotId = $snap.Id; DiskId = $newDisk.Id
  CloneVmId = (Get-AzVM -ResourceGroupName $TargetResourceGroup -Name $CloneVmName).Id
} | ConvertTo-Json | Out-File ".\$ChangeRef-clone-evidence.json"
Write-Host "Clone $CloneVmName created. Rename the OS before joining any domain." -ForegroundColor Green
```

EXAMPLE · RUNNING THE TEMPLATE

INVOCATION POWERSHELL

```
.\Clone-AzVm.ps1 -SubscriptionId '<sub-guid>' -SourceResourceGroup 'rg-app-prod' -SourceVmName 'APPVM01' `
  -TargetResourceGroup 'rg-app-test' -CloneVmName 'APPVM01-TEST' -VnetResourceGroup 'rg-net' `
  -VnetName 'vnet-test' -SubnetName 'snet-workload' -ChangeRef 'CHG0012345' -DiskSku StandardSSD_LRS -StopSource -WhatIf
```

Run first with `-WhatIf` to preview the destructive step (stopping the source), then run without it.

KEY CONCEPT · WHY `-CREATEOPTION ATTACH`?

When you build a VM from a marketplace image, the OS disk is created with `FromImage` and Azure runs first-boot provisioning (sets the admin password, hostname, etc.). With `Attach`, Azure boots the disk exactly as it is, so no provisioning runs. That is why the clone keeps the source's credentials and hostname.

CHAPTER 5

Method C, D and E · Other Ways to Get the Same Result

5.1 Method C · Azure CLI

The Azure CLI (`az`) performs the same three stages. It is pre-installed in Azure Cloud Shell and is often preferred in DevOps pipelines.

CLONE-VM.SH — BASH

```
RG_SRC=rg-app-prod; VM_SRC=APPVM01; RG_DST=rg-app-test; CLONE=APPVM01-TEST; CHG=CHG0012345
OSDISK_ID=$(az vm show -g $RG_SRC -n $VM_SRC --query storageProfile.osDisk.managedDisk.id -o tsv)
az snapshot create -g $RG_DST -n ${VM_SRC}-SNAPSHOT --source $OSDISK_ID --sku Standard_LRS --tags ChangeRef=$CHG
az disk create -g $RG_DST -n ${CLONE}-OSdisk --source ${VM_SRC}-SNAPSHOT --sku Premium_LRS \
   --public-network-access Disabled --network-access-policy DenyAll --tags ChangeRef=$CHG
az vm create -g $RG_DST -n $CLONE --attach-os-disk ${CLONE}-OSdisk --os-type Windows \
   --vnet-name vnet-test --subnet snet-workload --public-ip-address "" --nsg "" \
   --os-disk-delete-option Delete --nic-delete-option Delete --tags ChangeRef=$CHG
```

5.2 Method D · Azure Compute Gallery (generalized golden image)

When the goal is *many* VMs from one build, cloning disks one at a time does not scale and every copy shares a SID. Instead you **generalize** a reference VM and publish it as an image version in an **Azure Compute Gallery** (formerly Shared Image Gallery). Galleries version images, replicate them to other regions and can be shared across subscriptions.

1. Clone the source with Method A–C so you never sysprep the original (Sysprep is one-way).
2. Inside the clone run `C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown` (Linux: `sudo waagent -deprovision+user`).
3. In the portal, open the VM and select **Capture** › *Share image to Azure compute gallery* › *Generalized*. Or in PowerShell: `Set-AzVM -Generalized` then `New-AzGalleryImageVersion`.
4. Deploy new VMs from **Azure compute galleries › image › + Create VM**. Each gets new credentials and identity.

GALLERY CAPTURE (OUTLINE) — POWERSHELL

```
Stop-AzVM -ResourceGroupName rg-img -Name REF-CLONE -Force
Set-AzVM  -ResourceGroupName rg-img -Name REF-CLONE -Generalized
$vm = Get-AzVM -ResourceGroupName rg-img -Name REF-CLONE
New-AzGalleryImageVersion -ResourceGroupName rg-img -GalleryName galCorp -GalleryImageDefinitionName win2025-app `
  -Name 1.0.0 -Location eastus2 -SourceImageVMId $vm.Id -ReplicaCount 1
```

5.3 Method E · Restore from Azure Backup or a restore point

If the VM is protected by a Recovery Services vault, **Backup › Restore VM › Create new** builds a new VM from any retained recovery point. This gives an **application-consistent** copy "as of" a past time without touching the running source, which makes it ideal for disaster-recovery drills. VM *restore points* (`New-AzRestorePoint`) provide the same capability without a vault.

Method	Identity of result	Consistency	Speed	Typical use
A / B / C	Specialized (same SID)	Crash-consistent (clean if stopped)	Minutes	Test copy, upgrade rehearsal
D Gallery	Generalized (new SID)	Clean (sysprepped)	30–60 min first time	Fleet, scale sets, golden image
E Backup	Specialized	Application-consistent	15–60 min	DR drill, point-in-time copy

CHAPTER 6

Secure Access, Verification and Hardening

6.1 Reaching a VM that has no public IP

The clone deliberately has no public IP. Exposing RDP (3389) or SSH (22) to the internet is one of the most exploited misconfigurations in cloud environments and is prohibited by the CIS Azure benchmark. Use one of two sanctioned paths:

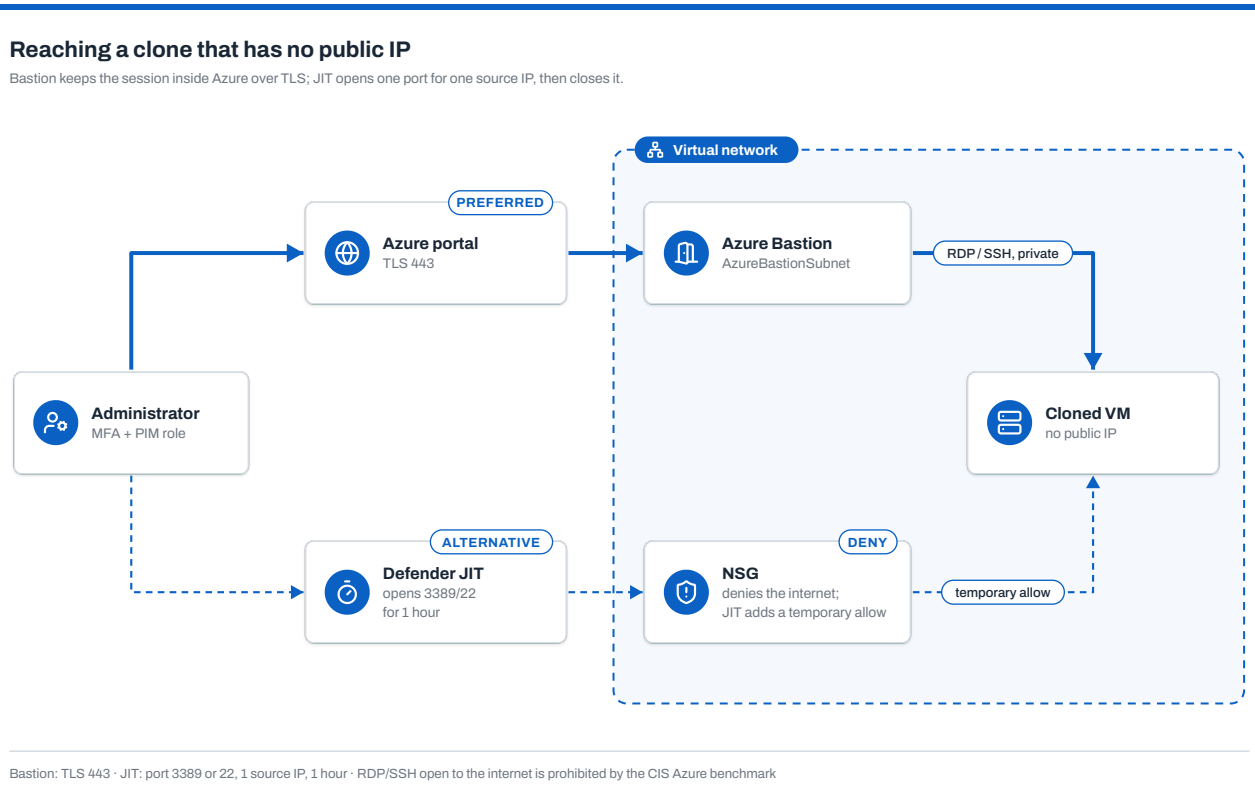


Figure 11. Prefer Bastion; use JIT only when you need a native client, and let it close the port for you.

- **Azure Bastion** (preferred): VM › Connect › Bastion, enter the source VM's local admin credentials. The session runs in your browser.
- **Just-In-Time access**: Defender for Cloud › Workload protections › Just-in-time VM access › Request access, choose *My IP*, 1 hour, port 3389.

6.2 Verification checklist

Azure side

- ☐ Status *Running*; size, zone, disk and NIC as planned
- ☐ No public IP on the VM or NIC
- ☐ Boot diagnostics screenshot shows the login screen
- ☐ Defender for Cloud lists the VM as protected
- ☐ Activity log shows snapshot, disk and VM create by the approved operator

Inside the guest

- ☐ OS version and patch level match the source
- ☐ All drives present with expected capacity
- ☐ Critical services start automatically
- ☐ Defender for Endpoint running (`Get-MpComputerStatus`)
- ☐ No new errors in System/Application event logs

6.3 Rename and harden before production use

INSIDE THE CLONE (ELEVATED) POWERSHELL

```
Rename-Computer -NewName 'APPVM01-TEST' -Restart
# If the source was domain-joined (run after the rename, on an isolated network first):
Remove-Computer -UnjoinDomainCredential (Get-Credential) -WorkgroupName TEMP -Restart -Force
Add-Computer -DomainName corp.example.com -Credential (Get-Credential) -Restart
```

```
# Rotate the inherited local admin password
Set-LocalUser -Name 'localadmin' -Password (Read-Host -AsSecureString 'New password')
```

Then update any application configuration that references the old hostname, onboard to Azure Update Manager, and delete the snapshot if it has no further value (snapshots bill until deleted).

CHAPTER 7

Rollback, Cost and Troubleshooting

7.1 Rollback by stage

Failure point	Rollback action
Stage 1 (snapshot)	Delete the partial snapshot; restart the source if you stopped it. No impact to production.
Stage 2 (disk)	Delete the failed disk; keep the snapshot only if retrying.
Stage 3 (VM)	Delete the VM. With *Delete with VM* enabled, the disk and NIC go too. Sweep the resource group for orphans (`Get-AzResource -TagName ChangeRef -TagValue CHG…`).
Post-deployment	Disconnect sessions, capture logs, delete the clone, open a problem record against the ChangeRef.

TAG-BASED CLEANUP POWERSHELL

```
Get-AzResource -TagName ChangeRef -TagValue 'CHG0012345' |
  Sort-Object {
@('Microsoft.Compute/virtualMachines','Microsoft.Network/networkInterfaces','Microsoft.Compute/disks','Microsoft.Compute/snapshots').IndexOf($_.ResourceTy
pe) } |
  Remove-AzResource -Force -WhatIf      # remove -WhatIf to execute
```

7.2 Cost considerations

- Snapshots and managed disks bill for provisioned size whether or not a VM is running.
- A deallocated VM stops compute charges only; disks and reserved IPs still bill.
- Defender for Servers Plan 2 bills per server from the first hour; Bastion has an hourly base charge shared across the VNet.
- Use B-series sizes, Standard SSD and auto-shutdown for non-production clones.

7.3 Troubleshooting

Symptom	Likely cause and fix
Clone will not boot (boot-diagnostics shows a recovery screen)	CMK disk encryption set unreachable in target region; or security type mismatch (Trusted Launch disk attached to a Standard VM). Match the source security profile.
Cannot connect via Bastion	NSG on the workload subnet must allow 3389/22 from the AzureBastionSubnet range.
JIT request denied	Defender for Servers Plan 2 not enabled, or requester lacks the `initiate/action` permission.
"Trust relationship failed" on source or clone	Both joined the domain with the same name/SID. Isolate, rename, and rejoin the clone; reset the source's secure channel with `Test-ComputerSecureChannel -Repair`.
Application errors on the clone	Hard-coded hostnames, IPs or connection strings. Search configs for the old name.

CHAPTER 8

Summary, Glossary and Review

8.1 Chapter summary

- An Azure VM is a set of resources; cloning means copying the OS disk and building a new VM around it.
- Snapshot → managed disk → VM produces a *specialized* copy with the same identity. Rename it before it meets the domain.
- Use PowerShell or CLI for repeatable, tagged, auditable clones; use a Compute Gallery for fleets; use Backup for point-in-time, application-consistent copies.
- Grant roles at the resource group, activate them just-in-time, and never give the clone a public IP.

8.2 Glossary

Managed disk	Azure-managed block storage volume attached to VMs.
Snapshot	Read-only point-in-time copy of a single managed disk.
Specialized image	Disk copy that retains machine identity, accounts and settings.
Generalized image	Disk with machine-specific identity removed by Sysprep or waagent.
SID	Security identifier; unique machine identity used by Windows and Active Directory.
Trusted Launch	VM security type adding Secure Boot, vTPM and boot-integrity monitoring.
Azure Bastion	Managed PaaS jump host providing browser RDP/SSH over TLS.
JIT VM access	Defender for Cloud feature that opens management ports on request for a limited time.
PIM	Privileged Identity Management; time-bound, approved activation of roles.

Check your understanding

1. Why can't you attach a snapshot directly to a new VM?
2. A clone of a domain-joined server is powered on in the production VNet before being renamed. What will break, and for which machine?
3. Which method would you choose to build ten identical web servers, and what must happen to the reference VM first?
4. Name the least-privilege roles needed to perform all three stages in a single resource group.
5. What is the difference between a crash-consistent and an application-consistent copy, and which methods produce each?